PHASE 7

USER EVALUATION FRAMEWORK

Comparison of a Static Explanation Baseline and an Interactive Power BI Dashboard

Master's Thesis Title:
Design and Evaluation of an Interactive Visual Analytics System for Explainable Artificial Intelligence

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1. Purpose of the Evaluation

The purpose of this phase is to assess whether the interactive Power BI dashboard supports a clearer and more useful interpretation of the customer churn prediction model than a fixed, non-interactive presentation.

The evaluation focuses on the practical use of the system. It examines whether users can understand churn predictions, identify important SHAP factors, compare customer groups, and make suitable customer-retention decisions using the information provided.

For this stage of the thesis, a user evaluation was conducted using ten participant profiles with different educational backgrounds and different levels of experience with Power BI.

The main aim is to compare two presentation formats:

- A static explanation baseline
- An interactive Power BI dashboard

Both systems use the same underlying model predictions, SHAP explanations, customer information, and model-performance results. The main difference is that the interactive dashboard allows the user to explore and filter the information.

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2. Systems Compared

2.1 Condition A — Static Explanation Baseline

The static explanation baseline presents fixed screenshots and visual summaries from the completed analytics system.

The static version includes:

- Model-performance metrics
- Global SHAP feature importance
- Customer churn probability

- Customer-level positive SHAP factors
- Customer-level negative SHAP factors
- Customer-risk information
- Customer-segment findings
- Classification-threshold information

Users can read and interpret the information shown in the static baseline. However, they cannot apply filters, change the selected customer, move dynamically between customer profiles, or interact with the visualizations.

The static baseline is intended to represent a conventional reporting format in which the available information is presented in a fixed form.

2.2 Condition B — Interactive Power BI Dashboard

The interactive condition uses the completed Power BI dashboard developed during Phase 6.

The report contains six pages:

1. Executive Overview
2. Customer Risk Analysis
3. Global Model Explanation
4. Individual Customer Explanation
5. Customer Segment Comparison
6. Model Performance Summary

Users can interact with the report by:

- Applying slicers and filters
- Selecting different customers
- Comparing customer segments
- Viewing dynamic churn probabilities
- Exploring global SHAP explanations
- Viewing individual SHAP explanations
- Navigating between report pages
- Reviewing model-performance and threshold information

The interactive dashboard is intended to support deeper exploration of the model results and explanations.

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3. Evaluation Objective

The main evaluation objective is to determine whether the interactive dashboard offers practical advantages over the static baseline.

The evaluation examines whether the interactive system improves:

- Understanding of the model prediction
- Identification of important churn factors
- Customer-retention decision quality
- User confidence
- Trust in the prediction system
- Ease of use
- Perceived usefulness
- Task accuracy

Task-completion time is also recorded to examine whether interactivity helps users find the required information efficiently.

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4. Research Questions

The evaluation is guided by the following research questions.

RQ1: Does the interactive Power BI dashboard improve users' understanding of customer churn predictions compared with the static explanation baseline?

RQ2: Does the interactive dashboard help users identify the most important global and customer-specific churn factors more accurately?

RQ3: Does the interactive dashboard support better decisions when identifying customers who may require retention attention?

RQ4: Does the interactive dashboard increase user confidence and trust in the model output and its explanations?

RQ5: Is the interactive dashboard perceived as more usable and useful than the static explanation baseline?

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5. Working Hypotheses

The following hypotheses were defined for the evaluation.

H1: Participants will achieve higher task accuracy when using the interactive dashboard.

H2: Participants will report a stronger understanding of the churn prediction and SHAP explanation information in the interactive condition.

H3: Trust and confidence ratings will be higher for the interactive dashboard.

H4: The interactive dashboard will receive higher usability and perceived-usefulness ratings.

H5: The interactive dashboard will provide better support for customer-retention decisions.

H6: The interactive dashboard may require additional exploration time, but it will support more complete and accurate answers.

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6. Evaluation Design

A within-subject comparison design is used for the evaluation.

This means that every participant profile evaluates both systems:

- The static explanation baseline
- The interactive Power BI dashboard

Using the same participant group for both conditions makes it possible to compare the systems directly.

Each participant completes equivalent tasks using both systems. The tasks cover the following areas:

- Global model interpretation
- Individual customer explanation
- Customer-risk analysis
- Customer-segment comparison
- Model-performance interpretation

To reduce the effect of system order, the participant profiles are divided into two groups.

Group 1

Participants P01 to P05 use the static explanation baseline first and the interactive dashboard second.

Group 2

Participants P06 to P10 use the interactive dashboard first and the static explanation baseline second.

This order arrangement helps reduce the possibility that the first system used influences the results of the second system.

Both conditions are based on the same underlying data and explanation outputs. The main difference between the conditions is the ability to interact with the information.

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7. Participant Profiles

The evaluation includes ten participant profiles.

The group contains participants from technical and non-technical educational and professional backgrounds. Their Power BI experience ranges from no previous experience to intermediate experience.

This variation makes it possible to examine how the dashboard may be understood by users with different levels of familiarity with data analytics and visualization tools.

Participant ID	Name	Age	Educational or Professional Background	Power BI Experience
P01	Lenin	23	Master's in Data Science	Basic
P02	Aahan	23	Pilot	None
P03	Steve	24	Master's in Cybersecurity	None
P04	Smith	24	Master's in Cybersecurity	None
P05	Nathen	24	Bachelor's in Data Science	Basic
P06	Deva	24	Master's in Data Science	Intermediate
P07	Deekshi	27	Master's in Architecture	None
P08	Helaine	23	Master's in Physics	None
P09	Akshata	23	Master's in Data Science	Intermediate
P10	Kshitija	23	Master's in Data Science	Intermediate

The participant profiles represent a range of backgrounds, including data science, cybersecurity, architecture, physics, aviation, and professional pilot training.

This mixture is useful because the dashboard is intended to communicate model information not only to machine learning specialists but also to users with limited technical knowledge.

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8. Evaluation Measures

The study records both task-performance measures and user-rating measures.

8.1 Task Accuracy

Task accuracy measures whether the participant provides the correct answer to each evaluation task.

Examples include:

- Identifying the most important global churn feature
- Finding the highest-risk customer

- Identifying the strongest customer-specific churn factor
- Selecting the customer group with the highest churn risk
- Interpreting the selected model threshold

8.2 Completion Time

Completion time records the time required to complete the task set for each system.

This measure helps determine whether users can locate and interpret the required information efficiently.

8.3 Understanding

Understanding measures how clearly the participant can interpret:

- The churn probability
- The predicted churn classification
- The SHAP feature contributions
- The global model behaviour
- The model-performance results

8.4 Trust

Trust measures whether the model prediction and explanation appear credible and understandable to the participant.

8.5 Decision Quality

Decision quality measures whether the participant can use the information to make a suitable customer-retention decision.

8.6 Usability

Usability measures how easy the system is to navigate, understand, and use.

8.7 Perceived Usefulness

Perceived usefulness measures whether the participant considers the system valuable for customer churn and retention analysis.

8.8 System Preference

After using both systems, each participant selects the system they prefer and provides a short reason for the preference.

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9. Rating Scale

Understanding, trust, decision quality, usability, and perceived usefulness are rated using a five-point Likert scale.

The scale is defined as follows:

1 — Strongly disagree
2 — Disagree
3 — Neutral
4 — Agree
5 — Strongly agree

The same rating scale is used for both systems to allow a direct comparison.

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10. Evaluation Procedure

Each participant profile follows the same general procedure.

1. Read a short explanation of the evaluation purpose.
2. Review the basic instructions for the first assigned system.
3. Use the first system and complete the assigned evaluation tasks.
4. Record the answers given for each task.
5. Record the total time required to complete the task set.
6. Complete the short system-rating questionnaire.
7. Review the instructions for the second system.
8. Use the second system and complete an equivalent task set.
9. Record the answers and completion time for the second condition.
10. Complete the second system-rating questionnaire.
11. Select the preferred system.
12. Provide one short comment about the main advantage or limitation of the systems.

The estimated time required for the complete evaluation is approximately 15 to 20 minutes per participant.

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11. Planned Data Collection

The following information will be recorded for each participant:

- Participant ID
- Participant name
- Age
- Educational or professional background
- Power BI experience
- System order
- Static-system task answers

- Interactive-system task answers
- Static-system task accuracy
- Interactive-system task accuracy
- Static-system completion time
- Interactive-system completion time
- Understanding ratings
- Trust ratings
- Decision-quality ratings
- Usability ratings
- Perceived-usefulness ratings
- Preferred system
- Short written feedback

The collected results will later be organised into a structured dataset for analysis.

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12. Planned Analysis

The final comparison will focus on the following results:

- Average task accuracy for the static baseline
- Average task accuracy for the interactive dashboard
- Average completion time for both systems
- Average understanding score
- Average trust score
- Average decision-quality score
- Average usability score
- Average perceived-usefulness score
- Number of participants preferring each system
- Common themes identified in participant comments

The results will be presented using summary tables and comparison charts.

Because the evaluation includes a limited number of participant profiles, the findings will mainly be interpreted as a practical demonstration of the proposed evaluation method.

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13. Expected Outcome

The interactive Power BI dashboard is expected to provide stronger support for exploring model predictions and SHAP explanations.

The interactive features may help users:

- Find information more easily
- Compare customer groups

- Change the selected customer
- Understand why a customer received a high churn probability
- Identify the most influential churn factors
- Make more informed retention decisions

The static baseline may be faster for simple questions because the information is fixed and immediately visible. However, it may provide less flexibility for deeper analysis.

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14. Phase 7.1 Summary

This section defined the evaluation framework for comparing the static explanation baseline with the interactive Power BI dashboard.

The following elements were established:

- Evaluation purpose
- Evaluation objective
- Systems being compared
- Research questions
- Working hypotheses
- Evaluation design
- Participant profiles
- Evaluation measures
- Rating scale
- Evaluation procedure
- Planned data collection
- Planned analysis